

ORIGINAL ARTICLE

Risk factors for the expansion of cutaneous leishmaniasis by *Leishmania tropica*: Possible implications for control programmes

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Abstract

Cutaneous leishmaniasis (CL) caused by *Leishmania tropica* is emerging in new areas, initially as outbreaks and then establishing endemic foci. There is little evidence of the risk factors and effectiveness of existing control measures, what limits our ability to generalize in different epidemiological settings. The disease is described as anthroponotic; however, zoonotic outbreaks have been reported in some countries. Our aim was to identify risk factors in a recently reported endemic focus in Morocco in order to design more effective control programmes. A case-control study was conducted from September 2014 to October 2015 for epidemiological data collection from families with and without CL cases. Sandflies were captured and *L. tropica* infection determined. The presence of potential animal reservoirs was evaluated. 71 CL cases (44 diagnosed between 2013 and 2015) and 137 healthy people were surveyed. The average age of the new cases was 33.1 ± 22.3 years, and 69.0% were women. *Phlebotomus sergenti* was the most abundant species with a density of 4.27 sandflies/trap/night and differences between houses with and without CL cases were detected (p -value = 0.014). Overall, 2.7% female *P. sergenti* and 3.0% dogs were positive for *L. tropica*. Human, cat, rabbit and bird blood was detected in blood-fed *P. sergenti* females. 45% people used preventive measures that were not translated into a reduction in the individual risk of acquiring CL. Exposure to *P. sergenti* was the only risk factor found, and the reduction in its density could be achieved through the improvement of water wells management, organic fertilizers' disposal and dogs control. The lack of effectiveness of indoor residual spraying and treated nets are attributable to poor compliance and misuse of them. In addition, result optimization of the awareness campaigns on the public is possible by involving patients with CL to explain their own experience.

KEYWORDS

control strategies, cutaneous leishmaniasis, *Leishmania tropica* expansion, *Phlebotomus sergenti*, potential animal reservoirs, risk factors

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1 | INTRODUCTION

Cutaneous leishmaniasis (CL) is a disfiguring and stigmatizing parasitic disease that is caused by a variety of *Leishmania* species. It is endemic in more than 70 countries with an incidence of 1.2 million cases per year (Alvar et al., 2012). Among these species, *Leishmania tropica* is the major cause of anthroponotic CL (ACL) in the Middle East and some areas of North Africa (Pratlong et al., 2009). The disease is described as anthroponotic; however, zoonotic outbreaks have been reported in some countries (Bousslimi, Ben-Ayed, Ben-Abda, Aoun, & Bouratbine, 2012; Svobodova et al., 2006). In the Middle East, civil war and terrorist conflicts are causing the largest human displacement in decades. There is a risk of outbreaks associated with the migration of populations affected with ACL. In the early 2000s, a CL outbreak was reported after the Iraq War that affected foreign troops in the area as well (Croft, Lestringant, & Baker, 2006; Du, Hotez, Al-Salem, & Acosta-Serrano, 2016). This disease has emerged or re-emerged in Turkey, Pakistan and Lebanon in association with the influx of refugees from Syria or Afghanistan (Du et al., 2016; Mubashir et al., 2017; Özkeklikçi, Karakuş, Özbel, & Töz, 2017). Knowledge of CL in conflict zones and among refugees is limited; nevertheless, the socioeconomic and environmental circumstances created by the conflict increase the risk factors for the continuous CL transmission and potentiate incidence (Du et al., 2016). According to Bailey et al. (2017) scarring or inactive CL is not included in prevalence estimates; however, the psychosocial consequences of past infection justify its inclusion in the global burden of CL disease estimates (Bailey et al., 2017). *Phlebotomus sergenti* is not only highly susceptible to *L. tropica* infection and a proven vector of this parasite, but also the main vector of ACL (Guilvard et al., 1991; Kamhawi, Modi, Pimenta, Rowton, & Sacks, 2000; Schnur et al., 2004). In Morocco, *P. sergenti* exhibited a wide ecological plasticity although it is believed to have a marked preference for semi-arid habitats (Boussaa, Pesson, & Boumezzough, 2007). In emerging Moroccan foci of *L. tropica*, the density of *P. sergenti* varies from 4 to 16 specimens/m² (Ramaoui, Guernaoui, & Boumezzough, 2008) with 12.8% to 76.7% of relative abundance (Boussaa et al., 2007), whereas a density of 1.79 has been reported in typical foci of ACL in Tunisia (Rioux et al., 1986).

Phlebotomus sergenti is present in some parts of southern Europe reaching figures of 35.27 *P. sergenti*/m² and representing up to 75% of total captures in certain sites (Barón et al., 2013); therefore, increased awareness is essential to prevent and control CL outbreaks. Nevertheless, there is little evidence of the risk factors and effectiveness of existing control measures of disease, what limits our ability to generalize in different epidemiological settings and cultures.

Recently, the World Health Organization classified 12 countries as “high-burden countries” for CL and several countries where *L. tropica* is endemic were selected, including Morocco (WHO, 2016). In Morocco, *L. tropica* has the widest geographic distribution and it is considered as a public health threat by the Ministry of

Health (Moroccan Health Ministry, 2011). Since 1989, this anthroponotic form of CL has been mainly reported in hypoendemic rural foci scattered around the subarid area of central Morocco. Later, CL caused by *L. tropica* emerged in several northern provinces of the country (Taza and Fes provinces) and in new areas of central and southern Morocco, initially as new outbreaks and then establishing endemic foci showing the expansion of this *Leishmania* species (Ajaoud et al., 2013).

In this way, the first CL case in Settât province (central Morocco) was detected in El Borouj locality in 2006, preceding an epidemic outbreak (Amarir et al., 2015). Currently, El Borouj is still the only active CL focus in the province of Settât, and it is considered endemic (Moroccan Health Ministry, 2011).

There are many potential risk factors for CL, and the aim of this study was to identify them specifically in El Borouj and to assess their relevance in order to design more effective control programmes. The control measures designed might not only serve for El Borouj, but also for other areas under similar epidemiological circumstances.

2 | MATERIALS AND METHODS

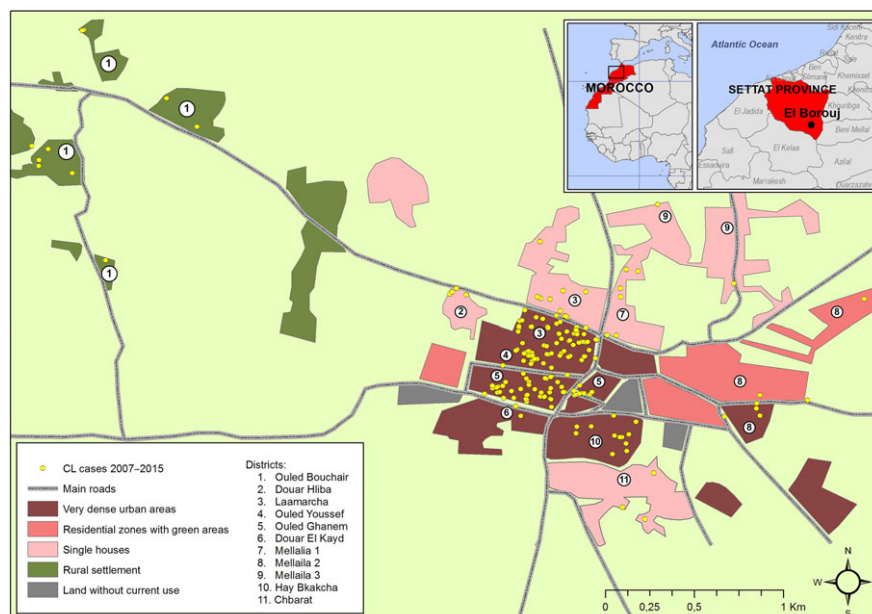
2.1 | Ethical issues

The procedure was approved by the Ethics Committee of the University of Granada and by the Moroccan Ministry of Health. Data on CL cases were derived from routine surveillance activities by the local healthcare centre and the delegation of the Ministry of Health in Settât Province (*Cellule provinciale de coordination*) with active and passive case detection. Once a year, visits were made by health professionals to schools and family home for the active detection of CL cases. The study and its procedures were explained to the mayor and heads of local healthcare centre and households. Written consent was obtained on a consent form from the enrolled families.

2.2 | Study area

The locality of El Borouj is situated at an altitude of 410 m above sea level (coordinates 07°36'W–32°29'N) in the Settât Province, central Morocco (Figure 1). Although it is classified as an urban municipality according to the national regional distribution, it still presents features of a rural area. The economic activity is mainly based on agriculture and animal breeding. The population is around 20,000 inhabitants, and its average growth rate is close to 2%. It is organized in three distinct zones according to their urbanistic evolution: classical “urban” neighbourhoods before the urbanistic restructuring, neighbourhoods that evolved from “douars” of rural character and neighbourhoods that evolved from slums. Differences in the systems of liquid and solid waste management, connection to the electric network and house dispersion were noticed; the absence of sanitation network was a characteristic in “douars” and slums.

FIGURE 1 Map of El Borouj showing the georeferenced cutaneous leishmaniasis cases confirmed by molecular and/or parasitological analysis from 2007 to 2015



In June 2006, the first case of CL in Settat province was identified in El Borouj locality: a 68-year-old woman who presented with a skin lesion on her nose. Subsequently, the number of autochthonous cases increased: Between 2007 and 2012, overall 356 CL cases were reported and *L. tropica* (MON-102 zymodeme) was identified as the aetiological agent (Amarir et al., 2015).

All the CL cases confirmed through molecular and/or parasitological analysis from 2007 to 2015 were georeferenced to perform a spatial and temporal analysis.

2.3 | New CL cases in El Borouj (2013–2015)

Microscopic diagnosis of new CL cases was performed at the health centre of El Borouj according to the protocol of the Moroccan Ministry of Health for active and passive case detection (Moroccan Health Ministry, 2011). Tissue scrapings were made at the edge of the lesions in suspected CL patients for detection of amastigote forms by microscopic examinations of stained slides. In some patients, a second slide was kept for molecular analysis. Confirmed CL patients received free intralesional injections of meglumine antimoniate according to the protocol of the Moroccan Ministry of Health and were covered with sanitary dressing. For some positive patients, *Leishmania* species identification was carried out with two different PCR techniques following the specifications described elsewhere or by manufacturers: ITS PCR-RFLP (Mouttaki et al., 2014) and Granaleish Multiplex qPCR (University of Granada, Spain, Trade Mark Number 3667362/5).

2.4 | Study design

A case–control study was conducted in El Borouj from September 2014 to October 2015 for the collection of epidemiological data. The case selection was based on the CL notification database from

the Ministry of Health Delegation in Settat province with the agreement of the concerned department at the Ministry of Health (Direction de l'Epidémiologie et de la Lutte contre les Maladies, Rabat). Cases occurred between October 2013 to October 2015 were randomly included. Cases from the 2007–2012 period were selected so that samples from every district were in the study. The following inclusion and exclusion criteria were established for the selection of the cases: Only autochthonous CL cases confirmed through molecular and/or parasitological analysis were considered, particularly the most recent ones reported between 2013 and October 2015, although previous cases were also included (Table 1). Imported cases that could have reached El Borouj from other areas were excluded. All family members of selected CL cases were also surveyed.

For each family with a CL case history included in the study, a neighbour family without any history of cutaneous lesion compatible with CL was chosen as control.

Epidemiological data of families with and without CL cases were collected through questionnaires. Demographic data, individual habits, travel history, housing condition and status of the immediate surroundings were recorded. We also asked about the knowledge of CL and protection methods used.

2.5 | Sandfly collection, morphological identification and *Leishmania* infection rate in the vector

Sandflies were caught in El Borouj using CDC light traps in the households of surveyed case/control families from September 20 to October 10, both 2014 and 2015. One to two traps were set in each house for one night with favourable weather conditions. Each case and its control were sampled at the same time using the same number of traps. Negative houses were sampled for a second night. The captured sandflies were placed in 70% alcohol. Male and female specimens were separated, and they were morphologically identified

TABLE 1 Individual demographic and clinical characteristics of the 44 CL cases diagnosed between October 2013 and October 2015

Demographic characteristics			Clinical characteristics		
Sex	Age	District	Year of occurrence	Number of lesion	Anatomical location of lesions
Female	67	Lamarcha	2014	3	Face
Female	51	Lamarcha	2014	1	Upper limbs
Male	46	Bkakcha	2015	1	Lower limbs
Female	11	Bkakcha	2015	1	Upper limbs
Male	3	Bkakcha	2015	1	Face
Male	17	Bkakcha	2014	2	Lower and upper limbs
Female	42	Bkakcha	2014	3	Upper limbs
Male	12	Bkakcha	2014	2	Face
Female	48	Bkakcha	2014	1	Face
Female	6	Bkakcha	2014	1	Face
Male	3	Bkakcha	2014	3	Face
Female	60	Bkakcha	2014	1	Face
Female	33	Bkakcha	2014	1	Face
Female	40	Bkakcha	2014	2	Face
Female	55	Bkakcha	2014	1	Upper limbs
Male	8	Bkakcha	2014	1	Face
Male	15	Bkakcha	2014	1	Face
Female	46	Bkakcha	2014	1	Face
Female	40	Bkakcha	2014	2	Face
Female	60	Bkakcha	2014	1	Upper limbs
Female	65	Bkakcha	2014	1	Lower limbs
Female	39	Bkakcha	2014	1	Upper limbs
Female	51	Bkakcha	2013	1	Upper limbs
Female	33	Ouled Bouchair	2015	3	Face
Female	16	Ouled Bouchair	2015	3	Face
Female	10	Ouled Bouchair	2015	1	Face
Female	9	Ouled Bouchair	2015	1	Face
Female	24	Ouled Bouchair	2014	4	Upper limbs
Female	19	Ouled Bouchair	2014	2	Upper limbs
Male	12	Ouled Bouchair	2014	1	Upper limbs
Male	12	Ouled Bouchair	2014	1	Upper limbs
Female	65	Ouled Bouchair	2014	1	Face
Female	59	Ouled Youssef	2014	1	Lower limbs
Female	60	Ouled Youssef	2014	1	Lower limbs

(Continues)

TABLE 1 (Continued)

Demographic characteristics			Clinical characteristics		
Sex	Age	District	Year of occurrence	Number of lesion	Anatomical location of lesions
Female	47	Mellalia	2015	2	Upper limbs
Female	10	Mellalia	2015	1	Face
Female	39	Mellalia	2014	1	Lower limbs
Female	10	Mellalia	2014	2	Face
Male	4	Mellalia	2014	1	Face
Male	13	Mellalia	2014	1	Face
Female	40	Douar Hliba	2014	2	Upper limbs
Female	60	Chbarat	2014	1	Upper limbs
Male	57	Douar El Kayd	2014	2	Lower limbs
Male	57	Douar El Kayd	2014	1	Upper limbs

using taxonomic keys. The head and genitalia of female specimens were individually removed for morphological identification, whereas the rest of the body was stored at -20°C for DNA extraction. Gonotrophic conditions of female sandflies were identified as blood-fed, nonfed or gravid. *Leishmania* species detection was carried out with two different PCR techniques: ITS PCR-RFLP (Mouttaki et al., 2014) and Granaleish Multiplex qPCR (University of Granada, Spain, Trade Mark Number 3667362/5).

2.6 | Blood meal identification

A PCR-RFLP method using universal primers complementary to the conserved region of the cytochrome b gene (Cytb) of the mitochondrion DNA (mtDNA) of vertebrates was applied for the identification of the meal source in sandflies; selective restriction enzyme cleavage of a short variable region of 359 bp with HaeIII and HinfI enzymes was performed (González et al., 2015). Blood samples from various potential hosts of different families were included as positive controls.

For comparative sequence analysis, the PCR product was purified and then directly sequenced in both directions using the same primers as for DNA amplification. The sequences were edited and aligned using Clustal-X 1.81 software. Subsequently, BLAST software was employed to determine similarities.

2.7 | Detection of the parasite in dog hair

Ear hair sample was extracted carefully with forceps from 33 dogs. Ten hairs from each dog were cut into small pieces with a sterile scalpel, placed in a sterile tube and grinded in liquid nitrogen with a pestle. DNA was isolated using the MasterPure[®] Extraction Kit (Epicentre, United States). *Leishmania* species detection was carried out with two different PCR techniques: ITS PCR-RFLP (Mouttaki et al., 2014) and Granaleish Multiplex qPCR (University of Granada, Spain, Trade Mark Number 3667362/5).

2.8 | Data analysis

ArcGIS 10.1 GIS software (ESRI, Redlands, CA, USA) was used for the design and construction of CL distribution maps.

The identification of the risk factors involved was performed through logistic regression (when the dependent variable was a categorical variable) and linear regression (when the dependent variable was a continuous variable) using the statistical package IBM SPSS Statistics 20. To normalize the density data and fulfil model assumptions, we used a natural logarithm transformation of $\ln(y + 1)$. A two-step procedure was used to select the most parsimonious model. First, all explanatory variables were assessed in a univariate model; variables significant at the $p \leq 0.2$ level were included in the final multivariate model (both, logistic or linear). Then, we used a stepwise backward elimination procedure: Variables with a p -value greater than 0.05 were removed.

3 | RESULTS

Georeferenced cases are showed at Figures 1 and 2; overall, there were 215 confirmed CL cases from 185 households.

All the cases that were analysed by molecular techniques (20 by Amarir et al., 2015 and 20 by us) were infected by *L. tropica*. In 14.1% of houses with CL history (26/185), between two and four CL cases had taken place, whereas in the rest of houses, only one CL case was found.

3.1 | Epidemiological data

Epidemiological data of 208 individuals (71 CL cases and 137 healthy people) living in 103 houses were collected through questionnaires. Forty-four of the CL cases were diagnosed between October 2013 and October 2015 (Table 1). In this endemic period, case diagnosis occurred between February and October; 45.5% of them were recorded in March and April, corresponding with the active search campaigns of CL cases of the Ministry of Health. The average age of cases was 33.1 ± 22.3 years old (from a minimum age of three to a maximum of 67 years); 47.5% of them were found in people younger than 20 years. Furthermore, 69.0% of the total number of cases were diagnosed in women but significant differences were only found in the age group above 20 years old (p -value= 0.007). The demographic characteristics and epidemiological data of the 71 CL cases collected through questionnaires are presented at Table 2.

3.2 | Sandfly collection and vector identification

Sandflies were captured in 96 households with a density of 8.44 specimens/trap/night. The vectors of the three *Leishmania* species responsible for CL in Morocco (*L. tropica*, *L. major* and *L. infantum*) were identified (Table 3). *Phlebotomus sergenti* was the most

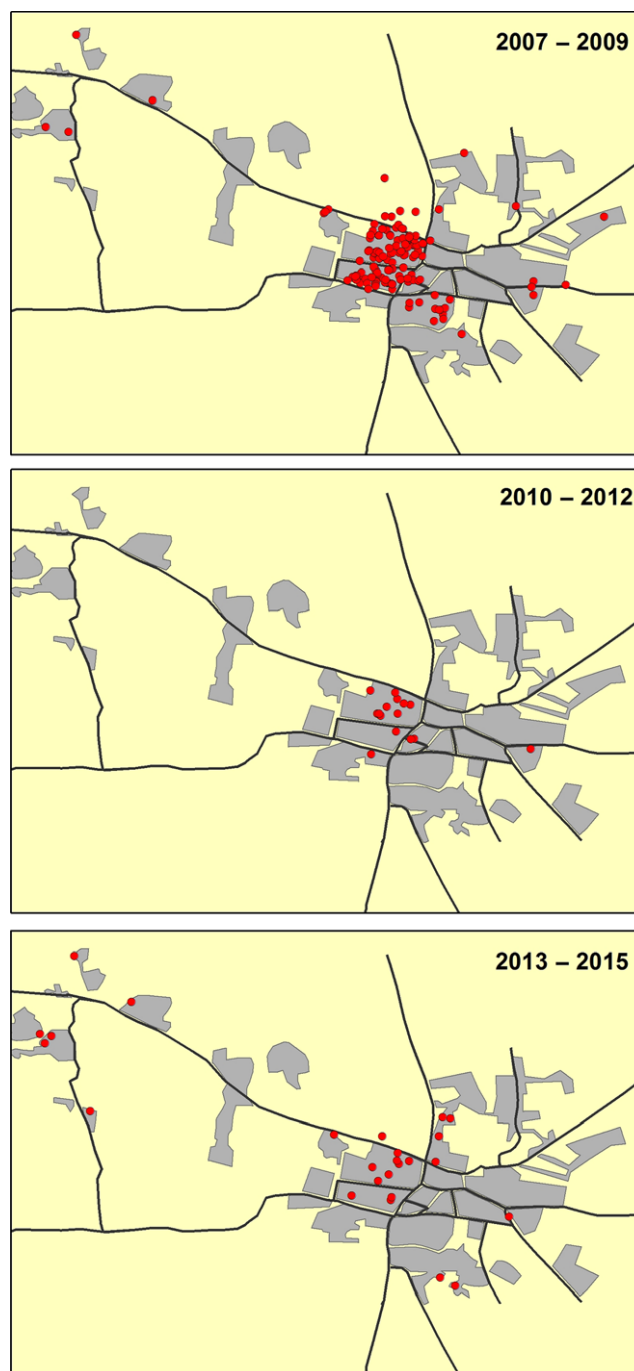


FIGURE 2 Maps of el Borouj showing the cutaneous leishmaniasis cases over time, from epidemic to endemic periods

abundant species (50.5%) with a density of 4.27 sandflies/trap/night and a predominance of females over males (Table 3). Although we captured *P. sergenti* in almost all houses, density was higher (p -value= 0.014) in the houses with CL cases diagnosed between 2013 and 2015 (5.77 sandflies/trap/night, CI95% 3.33–8.16) than in those not associated with CL cases (3.60 sandflies/trap/night, CI95% 1.98–5.24). Blood-fed, nonfed and gravid *P. sergenti* specimens were frequently found in the same household.

TABLE 2 Global characteristics of the 71 CL cases for which epidemiological data were collected through questionnaires

Age	32.9 ± 23.0, range: 3–80 years old
Sex	Male: 29 (40.8%) Female: 42 (59.2%)
Period of occurrence	2007–2012: 27 (38.0%) 2013–2015: 44 (62.0%)
District	Lamarcha 13 (18.3%) Bkakcha 22 (31.0%) Ouled Bouchair 9 (12.7%) Ouled Yousseff 10 (14.1%) Ouled Ghanem 3 (4.2%) Mellalia 7 (9.9%) Douar Hliba 1 (1.4%) Chbaratt 4 (5.6%) Douar El Kayd 2 (2.8%)
Number of lesions	1: 51 (71.8%) 2: 13 (18.3%) 3: 6 (8.5%) 4: 1 (1.4%)
Anatomical location of lesions	Face: 32 (45.1%) Upper limbs: 27 (38.0%) Lower limbs: 10: (14.1%) Upper and lower limbs: 2 (2.8%)
Type of lesion	Active: 14 (19.7%) Scare: 53 (74.6%) Both: 4 (5.6%)
Treatment with meglumine antimoniate	No: 3 (4.2%) Yes: 68 (95.8%)
Dressing covered lesions	No: 9 (12.7%) Yes: 62 (87.3%)
Knowledge of the disease	No: 13 (18.3%) Yes: 58 (81.7%)
Use of preventive measures	No: 38 (53.5%) Yes: 33 (46.5%)
Use of measures without scientific foundation	No: 61 (85.9%) Yes: 10 (14.1%)
Stay evenings at home	No: 50 (70.4%) Yes: 21 (29.6%)

3.3 | *Leishmania* infection rate in the vector

Five of 184 (2.7%) female *P. sergenti* tested for *Leishmania* infection were positive for *L. tropica* using molecular techniques. The five positive females were nonfed females and were collected exclusively in houses with CL cases.

3.4 | Potential animal reservoirs

The presence of potential animal reservoirs was evaluated through the analysis of *Leishmania* spp. in animal hair and the analysis of sandfly blood meals. *Leishmania tropica* was detected in one of 33 dogs (3.0%). For vertebrate DNA detection in female *P. sergenti*, a total of 49 engorged female *P. sergenti* were tested to determine the vertebrate host source of the blood meal. A positive Cytb PCR result was obtained for eight of the tested specimens and the origin of

blood meals was identified as *Homo sapiens* (three), *Oryctolagus cuniculus* (two), *Homo sapiens* + *Felis catus* (one) and *Emberiza sahari* (two). For the identification of the house bunting, it was necessary sequencing and comparative analysis of the Cytb fragment with a sequence library; a 97% of similarity was detected with *Emberiza sahari* sequence (JX515346.1).

3.5 | Analysis of risk factors associated with CL

Using an univariate logistic regression analysis, some of the factors not associated with CL (p -value >0.2) were the type of house (single-storey house or multistorey single-unit house), frequency of garbage collection, existence of a migrant in the family, indoor insecticide spraying and the use of mosquito nets (Table 4). Only the epidemiological data of the cases of the 2013–2015 period were considered in order to eliminate a possible bias due to the introduction of control measures after the appearance of the case. Risk factors associated with CL (p -value < 0.05) were the presence of livestock (keeping livestock inside or around the house), the proximity to a ruined house and the *P. sergenti* density. In the multivariate logistic regression model, only *P. sergenti* density was identified as a risk factor (p -value 0.014; Table 4). In addition, using a linear regression analysis *P. sergenti* density showed positive association with some districts, the presence of a water well, the accumulation of organic fertilizers around the house and the existence of dogs (Table 5).

4 | DISCUSSION

Since the beginning of the outbreak in El Borouj in 2007, CL cases were mainly concentrated in the most densely populated districts of the locality, which would support its anthroponotic nature. The epidemic outbreak lasted for 2 years, and since 2009, the number of CL cases began to acquire an endemic character (Figure 2). *Leishmania tropica* was the only *Leishmania* species identified by molecular techniques confirming previous results by Amarir et al. (2015) and according to Moroccan Health Ministry (2011).

In contrast to the epidemic period (Amarir et al., 2015), our results suggest that in the endemic period (2013–2015), there were no difference in the number of CL cases detected between different age groups. The predominance of CL cases in women compared to men usually described in Moroccan foci (Kahime, Laamrani-El Idrissi, Nhammi, & Boumezzough, 2016) was not present in the age group under 20 years. This higher rate of infection among women older than 20 years should still be attributed to the lack of medical care of men who give less importance than women and young people to the aesthetic appearance (Kahime et al., 2016) or are busier working to go to the health centre. On the other hand, even during the hot season, women stay evenings at home significantly more than men, as shown by the survey performed (odds ratio= 2.079, IC95% 1.065–4.056, p -value = 0.032), what might lead to a greater contact with the vector in the housings.

TABLE 3 Total number of collected specimens by CDC traps and *Phlebotomus* species identification

Genus	Subgenus	Species	Number of captured specimens			Density	Abundance	Frequency
			Male	Female	Total			
<i>Phlebotomus</i>	<i>Paraphlebotomus</i>	<i>Phlebotomus sergenti</i>	272	432	704	4.267	50.539	0.825
		<i>P. chabaudi</i>	0	1	1	0.006	0.072	0.010
	<i>Larroussius</i>	<i>P. longicuspis</i>	204	153	357	2.164	25.628	0.423
		<i>P. perniciosus</i>	90	26	116	0.703	8.327	0.268
		<i>P. langeroni</i>	2	0	2	0.012	0.144	0.021
	<i>Phlebotomus</i>	<i>P. papatasi</i>	50	69	119	0.721	8.543	0.433
		<i>P. bergeroti</i>	0	8	8	0.048	0.574	0.082
	<i>Phlebotomus spp</i>		1	0	1	0.006	0.072	0.010
<i>Sergentomyia</i>	<i>Sergentomyia spp</i>		48	37	85	0.515	6.102	0.392
Total			667	726	1393	8.442	100	0.866

Note. Density (specimens/trap/night), abundance (% specimens of a given species/total sandflies) and frequency (% positive houses for a given species) data were estimated.

Phlebotomus sergenti was the most prevalent species in the present work, similar to others emerging CL foci due to *Leishmania tropica* in semi-rural areas in central Morocco and showing similar *L. tropica* infection rates as well (Ajaoud et al., 2013).

In this study, we showed that differences in the exposure to the *L. tropica* vector (reflected by differences in *P. sergenti* density in the households) were the only factor associated with CL cases in the 2013–2015 period (Table 4). On the other hand, the presence of water wells, organic fertilizers and dogs are independently associated with a greater vector density in the household. The Ouled Ghanem district showed a significantly higher sandfly density as well (Table 5); even so, there have been no CL cases during the study period (Figure 2). Although vectors are present in nearly all the houses, improving management of water wells and organic fertilizers storage and disposal would reduce household vector density. The existence of damp sources has been identified as an important risk factor for *P. sergenti* presence (Barón et al., 2013). Coverage of the wells would reduce this moisture supply. None of the investigated factors was associated with the higher *P. sergenti* density in the Ouled Ghanem district (p -value > 0.2). However, it is the neighbourhood with the highest population growth in the last 20 years (annual growth rate of 2.6%) evolving from “douar” with rural character to densely populated urban neighbourhood. Several studies have highlighted the influence of environmental changes on the distribution and abundance/density of the vector sandfly species, and one of the most notable changes is the urbanization of land (Maroli, Feliciangeli, Bichaud, Charrel, & Gradoni, 2013).

In Morocco, the epidemiology of leishmaniasis caused by *L. tropica* has not been fully elucidated. CL due to this *Leishmania* species seems to increase and tends to extend into nonendemic areas, in contrast to CL due to *L. major* or *L. infantum* (Kahime et al., 2016). The disease is often described as anthroponotic and affects both rural and urban populations (Kahime et al., 2016; Rhajaoui et al., 2007). However, it has been suggested that it could be zoonotic in

some cases (Ajaoud et al., 2013; Guessous-Idrissi, Berrag, et al., 1997; Guessous-Idrissi, Chiheb, et al., 1997). Zoonotic outbreaks have been reported in other countries involving some rodent species as reservoirs, such as the rock hyrax in Israel, *Gundi* in Tunisia etc. (Bousslimi et al., 2012; Svobodova et al., 2006). Furthermore, in Morocco, *L. tropica* has been isolated from dogs but its role as reservoir was not considered (Dereure et al., 1991; Guessous-Idrissi, Berrag, et al., 1997; Guessous-Idrissi, Chiheb, et al., 1997; Lemrani, Nejjar, & Pratlong, 2002).

Analysis of dog hair is an optimal and noninvasive method for diagnosis and monitoring treatment response in canine leishmaniasis when molecular techniques are used (Corpas-López et al., 2016) and may be very useful in epidemiological studies (Belinchón-Lorenzo et al., 2013). In El Borouj, *L. tropica* was detected in 3.0% of the dogs through the PCR analysis of their hair.

Female sandflies need to take a blood meal for the protein supplementation essential for egg production. The study of the blood-feeding pattern of sandflies may help in the understanding of their interactions with potential reservoir hosts of *Leishmania* parasites. While the preference of certain sandflies species for a given animal or group (e.g., mammals or birds) may vary, the selective behaviour of some Phlebotomine species has been acknowledged. Nonetheless, most sandfly species for which information is available (*P. ariasi*, *P. perniciosus*, *P. perfiliewi*, *P. papatasi*, *Lutzomyia longipalpis*, etc.) appear to be generalists (González et al., 2015; Guy, Killick-Kendrick, Gill, Rioux, & Bray, 1984; Macedo-Silva et al., 2014; Maia et al., 2015) rather than specialists (*P. sergenti*, *P. longicuspis*) in their host range (Es-Sette, Ajaoud, Laamrani-Idrissi, Mellouki, & Lemrani, 2014). In this work, human, cat, rabbit and bird blood was detected in blood-fed *P. sergenti* females showing that *P. sergenti* fed not only on human but also on a wide range of animals. Feeding on various hosts species indicates an opportunistic behaviour of *P. sergenti*, what might influence *L. tropica* transmission in El Borouj area. Although CL cases

TABLE 4 Odds ratio for potential environmental risk factors of anthroponotic cutaneous leishmaniasis in El Borouj

Variable	Houses with CL cases N = 38	Houses without CL cases N = 46	Odds Ratio	Univariate 95%CI	p-Value	Odds Ratio	Multivariate 95% CI	p-Value
Type of house								
Single storey (ref)	12	17	1.265	0.507–3.158	0.873			
Multistorey	25	28						
Single house (ref)	7	10	1.190	0.404–3.508	0.752			
Grouped houses	30	36						
Wall condition								
unsuitable (holes) (ref)	20	24	0.982	0.415–2.321	0.967			
suitable	18	22						
Proximity to a ruined house								
No (ref)	24	17	0.3	0.140–0.833	0.018			
Yes	14	29	42					
Presence of vegetation								
No (ref)	18	18	0.643	0.266–1.553	0.326			
Yes	18	28						
Districts								
Laamarcha (ref)	2	5			0.840			
Hay Bkakcha	13	11	2.955	0.476–18.342	0.245			
Ouled Bouchair	7	6	2.917	0.407–20.899	0.287			
Ouled Youssef	2	6	0.833	0.084–8.240	0.876			
Ouled Ghanem	0	3	0.000	0.000	0.999			
Mellalia	6	4	3.750	0.473–29.752	0.211			
Douar Hliba	1	1	2.500	0.100–62.605	0.577			
Chbarat	5	6	2.083	0.275–15.772	0.477			
Douar El Kayd	2	4	1.250	0.118–13.240	0.853			
Presence of water well								
No (ref)	13	24	1.930	0.790–4.713	0.149			
Yes	23	22						
Accumulation of organic fertilizers								
No (ref)	31	43	2.312	0.514–10.402	0.275			
Yes	5	3						
Frequency of garbage collection								
daily (ref)	15	17	0.881	0.359–2.161	0.783			
irregular	21	27						
Presence of livestock								
No (ref)	22	37	2.616	0.000–7.039	0.050			
Yes	14	9						
Presence of dogs								
No (ref)	16	23	1.250	0.521–3.000	0.617			
Yes	20	23						
Presence of cats								
No (ref)	24	33	1.269	0.494–3.264	0.621			
Yes	12	13						
Presence of poultry								
No (ref)	18	26	1.300	0.542–3.120	0.557			
Yes	18	20						

(Continues)

TABLE 4 (Continued)

Variable	Houses with CL cases N = 38	Houses without CL cases N = 46	Odds Ratio	Univariate 95%CI	p-Value	Odds Ratio	Multivariate 95% CI	p-Value
Number of house occupants	6.2 (5.1–7.3)	5.6 (4.7–6.4)	1.085	0.934–1.260	0.284			
Existence of a migrant in the family								
No (ref)	14	17	0.954	0.387–2.349	0.919			
Yes	22	28						
Ln (Density of <i>P. sergenti</i> +1)	2.031 (1.657–2.404)	1.425 (1.097–1.753)	1.939	1.121–3.353	0.014	1.939	1.121–3.353	0.014
Curtains in the house								
No (ref)	21	29	1.688	0.594–4.798	0.326			
Yes	11	9						
Use of the fan								
No (ref)	28	35			0.983			
sometimes	5	7	0.893	0.256–3.118	0.859			
with high frequency	3	4	0.938	0.194–4.539	0.936			
Indoor insecticide spraying								
No (ref)	14	24	1.714	0.707–4.155	0.233			
Yes	22	22						
Use of mosquito nets								
No (ref)	29	35	0.768	0.264–2.234	0.628			
Yes	7	11						
Home control campaigns by health authorities								
No (ref)	12	19	1.655	0.653–4.193	0.288			
Yes	23	22						

Note. Only CL cases of the October 2013–October 2015 period were considered.

TABLE 5 Factors associated with *Phlebotomus sergenti* density predicted by the multivariate linear regression model

Factor		Univariate		Multivariate		Importance of the predictor factor ^a
		Coefficient	p-value	Coefficient	p-value	
Intercept				1.484	<0.001	
Districts	Laamarcha, Hay Bkakcha, Ouled Bouchair, Ouled Youssef	0.907	0.012	0.617	0.064	0.403
	Ouled Ghanem	1.909	<0.001	1.583	0.01	
	Mellalia, Douar Hliba, Chbarat	1.070	0.005	0.876	0.14	
	Douar El Kayd	Ref.		Ref.		
Intercept		0.299	0.378			
Water wells	Absence	−0.735	<0.001	−0.579	<0.001	0.393
	Presence	Ref.		Ref.		
Intercept		1.576	<0.001			
Organic fertilizers	Absence	−0.660	0.41	−0.593	0.039	0.128
	Presence	Ref.		Ref.		
Intercept		1.848	<0.001			
Dogs	Absence	−0.399	0.024	−0.250	0.05	0.076
	Presence	Ref.		Ref.		
Intercept		1.449	<0.001			

Notes. The continuous variable (natural) log (Density of *P. sergenti*+1) was used as the dependent variable for model construction. All houses for which sandflies were captured were considered: N = 96 houses, 51 with CL cases and 45 without CL cases. Intercept= mean (natural) log sandfly density of reference category. ^aIt represents the relative importance of each predictor in the estimate made by the model; its sum is 1.

are detected throughout the year, the dressing covering the lesions (Table 2) would difficult the vector infection, a fact that would supports the need of an animal host that contributes to the transmission (WHO, 2010).

On the other hand, the knowledge of the disease by the population of El Borouj was high (89.4%) and 45% people used preventive measures (indoor residual spraying of insecticides, curtains or treated nets) that were not translated into a reduction in the individual risk of acquiring CL during the 2013–2015 endemic period (p -value > 0.2). Generalized Indoor residual spraying with lambdacyhalothrin was conducted in 2007 to control transmission in the emerging focus. The insecticide susceptibility of *P. sergenti* was assessed during 2011 by Faraj et al., 2012, showing its susceptibility to the insecticides tested (DDT and lambdacyhalothrin) with a lower knockdown effect than that shown in areas where these insecticides have never been used for malaria or leishmaniasis control. Moreover, only 1.9% people spray insecticide in animal sheds usually found in the ground floor or in the terrace of the housing.

Reductions in CL incidence after the introduction of insecticide-treated bed nets have been demonstrated by diverse studies in Iranian, Turkish, Syrian and Afghanistan areas where *P. sergenti* was the dominant vector (Alten, Caglar, Simsek, Kaynas, & Perich, 2003; Emami, Yazdi, & Guillet, 2009; Jalouk, Al Ahmed, Gradoni, & Maroli, 2007). Its lack of effectiveness in Morocco has been attributed to poor compliance (Faraj et al., 2003, 2016).

In addition, 7.0% of the population applies measures of protection without scientific foundation like the use of lemon as a repellent. People who use measures without scientific basis are four times more likely to have CL than those who do not use it. Furthermore, the application of protection and control measures recommended by the sanitary authorities is three times greater in houses with a history of CL cases (p -value = 0.025), what reflects that the measures are applied after a CL case occurs in the family.

In conclusion, after 2 years of epidemic outbreak (2007–2008), El Borouj is currently an endemic focus of CL due to *L. tropica*. Our results support its anthroponotic nature, even though dogs might play a role in the maintenance of the parasite in this focus. Exposure to *P. sergenti* was the only risk factor for CL in the 2013–2015 endemic period, and the reduction in its density could be achieved through the improvement of management of water wells, organic fertilizers and dogs. The lack of effectiveness of indoor residual spraying and treated nets are attributable to both poor compliance and use of them. Educational tasks explaining the personal experiences of CL cases might improve the results of the awareness campaigns on the public, involving people in CL prevention and control.

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CONFLICT OF INTEREST

We declare no competing interests.

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